

Indeterminate forms — 0.9.8

Example: $\lim_{x \rightarrow 3} \frac{x-3}{x-3} = 1$ "indeterminate form of type $\frac{0}{0}$ "

$$\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3} = \lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3} = \lim_{x \rightarrow 3} x+1 = 4$$

Type $\frac{\infty}{\infty}$ $\lim_{x \rightarrow \infty} \frac{x^2 \rightarrow \infty}{x \rightarrow \infty} = \lim_{x \rightarrow \infty} x = \infty$

$$\lim_{x \rightarrow \infty} \frac{x}{x^2} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

Type $0 \cdot \infty$ $\lim_{x \rightarrow \infty} \frac{1}{x} \cdot x^2 = \infty$

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \underbrace{\cos(x)}_{\rightarrow 0} \underbrace{\tan(x)}_{\rightarrow \infty} = \lim_{x \rightarrow \frac{\pi}{2}^-} \cos(x) \cdot \frac{\sin(x)}{\cos(x)} = \lim_{x \rightarrow \frac{\pi}{2}^-} \sin(x) = 1$$

Type 0^0

$$\lim_{x \rightarrow 0} x^0 = 1, \quad \lim_{x \rightarrow 0^+} 0^x = 0$$

Type 1^∞

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e \approx 2.7$$

$$\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^x = e^{-1} = \frac{1}{e}, \quad \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^a$$

Type ∞^0

$$\lim_{x \rightarrow \infty} x^{\frac{1}{\ln(x)}} = \lim_{x \rightarrow \infty} \left(e^{\ln(x)}\right)^{\frac{1}{\ln(x)}} = \lim_{x \rightarrow \infty} e = e$$

$$\lim_{x \rightarrow \infty} x^{\frac{a}{\ln(x)}} = \lim_{x \rightarrow \infty} e^{\ln(x) \cdot \frac{a}{\ln(x)}} = \lim_{x \rightarrow \infty} e^a = e^a$$